Partial Derivatives and the Chain Rule

2026-08-17

MH2100 Calculus III

Week 03

Sources: MH2100_Lecture_03A, MH2100_Lecture_03B; Stewart Ch. 14, §§14.3–14.5

1 Where this sits

Week 02 ended with limits and continuity for $f : \mathbb{R}^2 \to \mathbb{R}$. This walkthrough covers both Week 03 lecture parts against the matching Stewart sections: partial derivatives (03A), then differentiability and the chain rule (03B).

2 Partial derivatives

Definition 2.1 (Partial derivative). For $f : \mathbb{R}^2 \to \mathbb{R}$,

$$f_x(a,b) \;=\; \lim_{h\to 0} \frac{f(a+h,\,b) - f(a,b)}{h},$$

provided the limit exists, and $f_y(a,b)$ symmetrically. Notation: $f_x = \partial f/\partial x = \partial_x f = D_1 f$. Higher partials iterate: $f_{xy} = (f_x)_y = \partial^2 f/\partial y\,\partial x$ — note the order flip between the subscript and the ∂ notation.

Example 2.2. For $f(x,y) = x^3 y + e^{xy}$: hold y fixed and differentiate in x, and vice versa,

$$f_x = 3x^2 y + y\,e^{xy}, \qquad f_y = x^3 + x\,e^{xy}.$$

Geometrically, $f_x(a,b)$ is the slope at $x = a$ of the trace $z = f(x,b)$ — a single-variable derivative along a slice, which is why every one-variable rule transfers unchanged.

Theorem 2.3 (Clairaut). *If f_{xy} and f_{yx} are both continuous on an open set containing (a,b), then $f_{xy}(a,b) = f_{yx}(a,b)$.*

Remark 2.4. The continuity hypothesis does real work. For $f(x,y) = xy\,(x^2 - y^2)/(x^2 + y^2)$ with $f(0,0) = 0$ (Stewart §14.3, ex. 103), $f_{xy}(0,0) = -1 \neq 1 = f_{yx}(0,0)$.

3 Differentiability

Definition 3.1 (Differentiable). f is *differentiable* at (a,b) if the increment decomposes as

$$f(a+\Delta x,\, b+\Delta y) - f(a,b) \;=\; f_x(a,b)\,\Delta x + f_y(a,b)\,\Delta y + \varepsilon_1\,\Delta x + \varepsilon_2\,\Delta y,$$

where $\varepsilon_1, \varepsilon_2 \to 0$ as $(\Delta x, \Delta y) \to (0,0)$: the tangent plane is a genuinely first-order-accurate approximation, not just a plane through the point.

Warning 3.2. Existence of both partials does *not* give differentiability — not even continuity. For $f(x,y) = xy/(x^2+y^2)$ with $f(0,0) = 0$, both partials exist at the origin ($f_x(0,0) = f_y(0,0) = 0$, since f vanishes on both axes), yet $f \equiv \frac{1}{2}$ along the line $y = x$, so f is not continuous there. The usable criterion is the sufficiency theorem: f_x, f_y exist near (a,b) and are continuous at $(a,b) \Rightarrow f$ differentiable at (a,b).

4 The chain rule

Theorem 4.1 (Chain rule, case 1). *If $z = f(x,y)$ is differentiable and $x = g(t)$, $y = h(t)$ are differentiable, then*

$$\frac{dz}{dt} \;=\; \frac{\partial z}{\partial x}\frac{dx}{dt} + \frac{\partial z}{\partial y}\frac{dy}{dt}.$$

Theorem 4.2 (Chain rule, general). *If $u = f(x_1,\dots,x_n)$ is differentiable and each $x_i = x_i(t_1,\dots,t_m)$ has continuous partials, then for each j,*

$$\frac{\partial u}{\partial t_j} \;=\; \sum_{i=1}^{n} \frac{\partial u}{\partial x_i}\,\frac{\partial x_i}{\partial t_j}.$$

Example 4.3. $z = x^2 y + 3xy^4$ with $x = \sin 2t$, $y = \cos t$; find dz/dt at $t = 0$. By case 1,

$$\frac{dz}{dt} = (2xy + 3y^4)(2\cos 2t) + (x^2 + 12xy^3)(-\sin t).$$

At $t = 0$: $x = 0$, $y = 1$, so $dz/dt = (0+3)(2) + (0+0)(0) = 6$.

Remark 4.4. The lecture's tree diagram is bookkeeping for the general theorem: one summand per path from u down to the target variable t_j, multiplying the derivatives along the path's edges.

Check yourself.

1. Rebuild the $xy/(x^2+y^2)$ counterexample unaided: compute both partials at the origin from the limit definition, then exhibit the discontinuity along $y = x$.
2. Stewart §14.5, exercises 1–12 shape: pick two, compute dz/dt both by substitution and by the chain rule, and confirm they agree.